

Umer Farooq

HPC, QUANTUM SIMULATION, AND AI SYSTEMS ENGINEER

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SUMMARY

Systems-focused computer scientist with deep experience in high-performance computing, parallel and distributed systems, and full-stack development. I build efficient, scalable systems that span GPU-accelerated algorithms, quantum simulation, multi-agent RAG pipelines, custom languages, distributed hash tables, and real-time networked applications.

EXPERIENCE

Software engineer

Sep 2025 – present

Open Quantum Workbench, FAST-NUCES · Islamabad, Pakistan

- Build the browser-side simulation interface and the Python compute services behind it, so a circuit can be run and inspected without a local quantum toolchain.
- Put Qiskit, Cirq, and PennyLane behind one interface, with OpenQASM 3.0 as the exchange format between them.
- Turn requirements from the group's researchers into APIs and UI that keep the physics legible.

Python · FastAPI · React · Qiskit · PennyLane · OpenQASM 3.0

Freelance developer — Level 2 seller

Aug 2023 – Aug 2024

Fiverr · Remote

- Delivered 30+ full-stack applications on the MERN stack and .NET, reaching roughly 500 end users.
- Wrote 2D games in C++ with SFML and SDL2 — custom physics, collision detection, and a rendering path 30% faster than the first working version.
- Built desktop line-of-business applications in C# and Java over PostgreSQL and MySQL, MVC throughout, sized for 100+ concurrent users.
- Closed 100+ projects at 98% client satisfaction and 80% repeat custom, delivering about 20% ahead of the agreed date.

React · Node.js · Express · MongoDB · PostgreSQL · C++ · C# · Java · .NET · SFML · SDL2

Core team member — PR and marketing

Feb – May 2025

NaSCon'25, FAST-NUCES · Islamabad, Pakistan

- Managed community partners at 15+ universities: inter-university meetings, RSVP tracking, and campaign content across LinkedIn, Instagram, and Facebook.
- Ran partner-facing promotion for one of Pakistan's largest student-run technology conferences.

Officer — communications and outreach

Feb – May 2025

NaSCon'25, FAST-NUCES · Islamabad, Pakistan

- Handled correspondence between FAST-NUCES Islamabad and partner universities, and confirmed speaker and sponsor invitations.

EDUCATION

Bachelor of Computer Science

Aug 2022 – Jun 2026 (Expected)

National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad Campus

- Dean's List Award – Spring 2023

COURSEWORK

High-Performance Computing · Parallel & Distributed Computing · Compiler Construction · Operating Systems · Advanced Programming · Data Structures & Algorithms · Database Systems · Computer Networks · Artificial Intelligence · Machine Learning

SELECTED PROJECTS · 8

QCanvas

Mar 2025 – Present

Unified Quantum Simulator Platform

3rd Prize · Huawei ICT Competition National Finals (UniQ Team)

Next.js · TypeScript · FastAPI · Qiskit · Cirq · PennyLane · OpenQASM

Cirq-RAG Code Assistant

Generative AI & RAG

Quantum code generation without hallucinations

CIRCUIT GENERATION SUCCESS 92% from 52% KNOWLEDGE BASE ENTRIES 140+

Python · Cirq · RAG · FAISS · Multi-Agent · Prompt Engineering

Parallel Tensor Network Quantum Simulator (Q-Tensor)

Academic/Research

Hybrid MPI/OpenMP + CUDA Quantum Simulation

C++ · CUDA · OpenMP · MPI · METIS · CMake · Quantum Computing

MNIST Classification with GPU Acceleration

High-Performance Computing & GPU

Deep Learning Optimization

CUDA · Python · PyTorch · Nsight · Tensor Cores · FP16 Mixed Precision

GPU-Accelerated Canny Edge Detection

High-Performance Computing & GPU

Image Processing Optimization

C++ · CUDA · OpenCV · Image Processing

Optimized RNN Character-Level Text Generation

Deep Learning & NLP

Shakespeare Text Generation

Python · PyTorch · CUDA · LSTM · NLP

Parallel Graph & Text Analysis

High-Performance Computing

Multi-threading with pthreads

C++ · pthreads · perf · Graph Algorithms

PixelRNN Implementation

Generative AI

CIFAR-10 Image Generation

Python · TensorFlow · Keras · CUDA · Generative AI

22 further projects — compilers, distributed hash tables, desktop applications, and games — are listed in full on the work page.

ALL 30 PROJECTS →

SKILLS

HIGH-PERFORMANCE COMPUTING & PARALLELISM

- GPU computing with CUDA, Tensor Cores, mixed-precision training
- OpenMP, MPI, and hybrid MPI+OpenMP parallel programming
- Performance profiling and optimization with NVIDIA Nsight, perf, and Valgrind

MACHINE LEARNING, DEEP LEARNING & LLM SYSTEMS

- PyTorch, TensorFlow/Keras for CNNs, RNNs/LSTMs, diffusion and GAN models
- Retrieval-Augmented Generation (RAG), semantic search, FAISS, evaluation with NDCG/MAP
- Multi-agent AI workflows, prompt engineering, and tool-augmented pipelines

QUANTUM COMPUTING & SIMULATION

- Qiskit, Cirq, PennyLane, and OpenQASM 3.0
- Quantum circuit simulation, tensor networks, and hybrid classical-quantum workflows

SYSTEMS, DISTRIBUTED & NETWORKING

- Modern C/C++ for systems programming and algorithmic optimization
- Distributed hash tables, IPFS-style storage, P2P and client-server architectures
- Socket programming (TCP/UDP), multithreading with pthreads, concurrency control

FULL-STACK DEVELOPMENT & PLATFORMS

- React, Next.js, Node.js/Express, FastAPI, RESTful API design
- .NET, JavaFX, desktop applications with PostgreSQL/MySQL backends
- Kubernetes, Docker, GitHub Actions, and CI/CD for reproducible deployments

CERTIFICATIONS

- 2025 Oracle Cloud Infrastructure Certified Generative AI Professional Oracle CREDENTIAL
- 2025 Oracle Cloud Infrastructure Certified AI Foundations Associate Oracle CREDENTIAL

AWARDS

2024 3rd prize — Huawei ICT Competition, national finals With the UniQ team, for QCanvas.

2023 Dean's List Spring 2023, FAST-NUCES.

2024 Level 2 seller, Fiverr 100+ projects delivered at 98% satisfaction.

LANGUAGES

Urdu NATIVE OR BILINGUAL English NATIVE OR BILINGUAL Japanese ELEMENTARY

Umer Farooq · <https://umer-farooq-cs.github.io/portfolio/cv>